| INCH-POUND |

MIL-C-82086C
18 February 1989
SUPERSEDING
MIL-C-82086B(1)
22 April 1983

MILITARY SPECIFICATION

COMPRESSOR, AIR, DIESEL-ENGINE-DRIVEN, PORTABLE, WHEEL-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial, portable wheel mounted, diesel-engine-driven air compressors of rotary vane or rotary screw design.

1.2 Classification. The compressors are of the following classes and sizes, as specified (see 6.2).

1.2.1 Class. The class of compressors consist of the following:

a. Class A - Wheeled undercarriage with fixed tongue.
b. Class B - Wheeled undercarriage with automotive steering and hinged towbar.

1.2.2 Size. The size of compressor consist of the following:

a. Size 125 - 125 cubic feet per minute (cfm) at 100 pounds per square inch gage (psig).
b. Size 250 - 250 cfm at 100 psig.
c. Size 365 - 365 cfm at 100 psig.
d. Size 600 - 600 cfm at 100 psig.
e. Size 750 - 750 cfm at 100 psig.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 4310

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

WW-C-633 - Coupling Halves, Cap and Wye; Quick Disconnect Pneumatic Hose, Two-Lug Universal Type.

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating Power-Driven; Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-C-83286 - Coating Urethane, Aliphatic Isocyanate, For Aerospace Applications.

STANDARDS

MILITARY

MIL-STD-209 - Sling and Tiedown Provisions for Lifitng and Tying Down Military Equipment.
MS51336 - Lunette-Coupler, Drawbar, Ring.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF THE AIR FORCE

DRAWING

7545352 - Requirement for Finishes, Protective, and Codes for San Antonio ALC Ground and Ground Support Equipment.

(Application for copies should be addressed to SA-ALC/MMEDO, Kelly AFB, TX 78241-5990.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 40, CFR, Part 204 Subpart B - Noise Emission Standards for Construction Equipment.

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.
Section VIII, Division 1 - Pressure Vessels.

ASME Power Test Code, PTC-9 - Displacement Compressors, Vacuum Pumps and Blowers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The air compressor shall consist of a diesel engine, air end, receiver, air pressure regulating system, lubricating and cooling systems, safety controls, housing, and the piping, instruments, gages, and other accessories specified herein all assembled and mounted on a fully suspended wheel-mounted undercarriage.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2.1 and 6.6).

3.3 Standard commercial product. The compressor shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the compressor being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 Federal Motor Vehicle Safety Standards. Wheeled undercarriages and furnished accessories shall comply with all applicable Federal Motor Vehicle Safety Standards (FMVSS) in effect on the date of manufacture.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using

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materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All compressors of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts. Air end components are not included in the interchageability requirements of this specification. No deviations shall be acceptable without written approval of the contracting officer.

3.6 Performance. The compressor shall be capable of compressing and delivering not less than the nominal volume of free air applicable to the compressor size specified at a receiver pressure of 100 psig under the following atmospheric conditions:

a. 14.696 pounds per square inch (psi) absolute (sea level).
b. 60 degrees Fahrenheit (°F).
c. Zero humidity.

Under the conditions specified, the air discharge temperature shall not exceed 250°F. The compressor shall be capable of operating satisfactorily when tilted 15 degrees (°) longitudinally or transversely from a level position.

3.6.1 Regulation. Regulation of compressor output shall be accomplished by a stepless, pneumatic control or electronic system. The system shall be adjustable within limits established by the manufacturer and shall be set to automatically maintain a nominal discharge pressure of 100 psig when the compressor is operating at full rated capacity. The system shall also provide for automatic regulation of compressor output from zero to 100 percent of rated capacity. Primary regulation shall be effected by the proportional and automatic reduction of engine speed within the range of full-rated-capacity speed to a speed not less than the engine manufacturer's recommended idle speed. Supplemental control may be accomplished by the proportional and automatic actuation of a compressor inlet unloader valve.

3.6.2 Environmental conditions. The compressor shall be capable of starting and operating under the following atmospheric conditions:

a. Plus 125°F to -20°F at sea level (29.92 inches of mercury).
b. Plus 102°F to -20°F at an elevation of 5,000 feet (24.98 inches of mercury).

3.6.3 Safety shutdown. The compressor shall be equipped with a safety shutdown system designed to automatically stop the engine under any of the following conditions:

a. Compressor discharge air temperature in excess of safe limit established by the manufacturer but in no case exceeding 250°F.

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b. Engine coolant temperature in excess of manufacturer's established safe limit.
c. Engine lubricating oil pressure below minimum manufacturer's established safe limit.

3.6.4 Sound level. The exterior sound level produced by the compressor shall not exceed the noise limits established by the EPA, Title 40, CFR, Part 204, Subpart B, in effect on the date of manufacture.

3.7 Engine. The air end shall be driven by a liquid- or air-cooled diesel engine of a size and type used or recommended for use by the compressor manufacturer on standard commercial compressors. The engine, at full-rated compressor output, shall have an intermittent horsepower (hp) rating sufficient to properly and adequately operate the air end and all connected auxiliaries at the speed required to develop rated compressor capacity under the atmospheric conditions specified in 3.6. Instruments and gages specified in 3.7.1 shall be of the type recommended by the manufacturer. All engine accessories supplied commercially as standard equipment shall be furnished. The engine shall be equipped with an electrical starting system rated either at 12 volts (V), or 24V in accordance with the standard voltage for the engine being furnished. The diesel engine shall be capable of starting within five minutes and shall be ready for full-load operation within 15 minutes under any of the environmental conditions specified in 3.6.2. The fuel tank shall be sized for continuous operation at rated compressor output of at least eight hours. The diesel engine shall start in any temperature above -20°F. Starting aids may be either electric glow plug or ether primer. When an ether priming system is required, it shall be the measured shot type, with storage capacity of not less than 12 fluid ounces. A compressor of the enbloc design is excluded from this specification.

3.7.1 Instruments and controls. The following engine instruments and controls, as a minimum, shall be furnished:

a. Control, starter.
b. Control, shutoff.
c. Control, manual engine warm-up.
d. Gage, lubricating oil pressure.
e. Gage, engine coolant temperature.
f. Ammeter or battery charging indicator.
g. Safety controls (see 3.6.3).
h. Priming aid controls (see 3.7).
i. Hour meter.

All instruments shall be mounted on a panel or panels located near the operating controls and shall be visible to the operator. The instrument panel(s) shall be furnished with illumination for night operation. The engine hour meter shall have a totalizing mechanism capable of registering not less than 9,999.9 hours of engine operating time.

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3.8 Storage battery(s). Unless otherwise specified (see 6.2), the manufacturer's standard commercial storage battery(s) normally furnished to make a complete electrical system shall be furnished dry charged without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture. When specified (see 6.2), a wet charged or wet sealed maintenance free battery(s) shall be furnished. The wet charged battery(s) shall be filled with electrolyte and fully charged. Electrolyte shall have a specific gravity of 1.265 ± 0.010 at 80°F.

3.9 Air cleaners. Air cleaners shall be of the two-stage dry-type design wherein the first stage provides centrifugal separation and the second stage provides impingement filtration with use of a suitable filter medium. When handling dust concentration of 0.025 grams per cubic foot, the air cleaners shall be capable of operating not less than 8 hours without exceeding an air restriction which could adversely affect the rated output of the compressor. When located outside the compressor housing, air intake shall be fitted with a rain cap and screen. Connections of components associated with the air cleaner installation shall be water- and gas-tight. The cleaner shall have an indicator that shows when the cartridge is ready for replacement and shall be visible to the operator when the enclosure door is open.

3.10 Radiators/heat exchangers. Radiators/heat exchangers shall be configured so that they may be cleaned from both sides with compressed air or water spray without disconnecting the associated hoses and piping. Cleaning of radiators/heat exchangers shall be completed in less than one hour.

3.11 Air end. Unless otherwise specified (see 6.2), the air end shall be of the rotary vane or rotary screw design.

3.12 Receiver. The air receiver shall be designed, constructed, and stamped for a pressure of not less than 150 psig, in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII, Division 1. The receiver shall be equipped with a pressure gage and drain valve or plug and shall be protected against excessive pressure by the manufacturer's standard relief valve. The receiver discharge system shall include a manual blowdown valve and an automatic blowdown valve or sonic orifice in the discharge designed to blow the complete air system down to atmospheric pressure when the engine stops. When a pressure in excess of 10 psig must be maintained in the receiver to insure adequate circulation of compressor oil of elimination of carry over, the air discharge piping shall include a minimum pressure valve or device set to maintain a predetemined minimum back-pressure on the compressor oil system.

3.13 Manifold and service outlets. The air receiver discharge line shall be equipped with an air service manifold having individually controlled service outlets with standard universal type pneumatic hose couplings. The air service manifolds of each size of compressor shall be fitted with not less than the number and sizes of outlets specified in table I. All 3/4-inch outlets shall be equipped with the manufacturer's standard lever-type air

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cocks. Couplings conforming to WW-C-633, type II, or type III, shall be provided. The compressor shall be equipped with a check valve installed in the air line between the receiver and the manifold outlets to prevent reverse flow of air into the receiver during blowdown. When specified (see 6.2), a manual shutoff valve shall be installed in the air line between the receiver and the manifold.

TABLE I. Service outlets.

Compressor size	Number of 3/4-inch outlets	Number of 1-1/2-inch outlets with gate, globe or plug valve
125	2	-
250	4	1
365	4	1
600	4	2
750	4	2

3.14 Housing. A suitable housing shall be provided to enclose the engine, compressor, and instrument panel(s). Housing and panels shall be not less than 0.0568 inch (U. S. revised standard gage No. 16) thick. Supporting members and braces shall be furnished as required. The housing shall be sufficiently rigid to support a 300-pound load applied to any single one-square-foot area on the top of the housing. The housing shall include hinged access openings to the fuel service, engine, air end, and instrument panel(s). Access panel(s) shall be equipped with devices to hold the panel(s) in an open position and clamps or latches to retain the panel(s) in the closed position. Access panel(s) to fuel service and instrument panels shall be lockable with a padlock. Fenders shall be sufficiently supported to withstand the load of a person weighing not less than 175 pounds (lbs).

3.15 Mounting. The complete air compressor shall be securely mounted on a wheel-mounted undercarriage equipped with parking brakes. Compressors shall be equipped with a wheeled or footed landing leg and drawbar lunette-coupler. Drawbar couplers shall conform to MS51336. All compressors shall be equipped with suspension components with a rated capacity at least equal to the loads imposed on each member, measured at the ground, with compressor loaded to capacity. When rating suspension capacity, unsprung weight shall be deducted. Two safety chains shall be furnished on each compressor. Each compressor shall be designed for towing over graded gravel roads at speeds up to 20 miles per hour (mph), and paved surfaces at speeds up to 50 mph.

3.15.1 Class A. The complete air compressor shall be securely mounted on a wheel-mounted undercarriage with a fixed towing tongue. When specified (see 6.2), the undercarriage shall be equipped with a towing tongue having a

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minimum adjustable hitch height of ±3 inches from the centerline of the tongue for size 125, and -3 inches to +11 inches from the centerline of the tongue for sizes 250 and greater. Size 125 shall be of the two-wheel single axle type. Sizes 250, 365, and 600 shall be of the four-wheel type with tandem axle. Size 750 shall be of the four-wheel or six-wheel type with tandem axles. The size 750 tandem undercarriage shall be equipped with surge-type brakes and shall incorporate a positive mechanical lockout to prevent brakes from locking when the compressor is pushed rearward by the towing vehicle.

3.15.2 Class B. The complete compressor unit shall be securely mounted on a wheel-mounted undercarriage with automotive type steering and hinged towbar. Sizes 365, 600, and 750 shall be of the four-wheel type.

3.15.3 Tires and wheels. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground with compressor loaded to rated gross vehicle weight. Tires shall not drag on fenders when making turns. Unless otherwise specified (see 6.2), tire size for 250 and 365 compressors shall be not less than 10:00 X 15; tire size for 600 and 750 compressors shall be not less than 12:00 X 15.

3.15.4 Parking brake. Unless otherwise specified (6.2), a lever type handbrake shall be provided. The parking brake shall be capable of holding the compressor in a stationary position on a 15 percent grade. Frictional parking brakes utilizing the exterior surface of the tires are not acceptable.

3.16 Air transportability. The compressor dimensions shall not exceed 105 inches in width and 102 inches in height in a reduced configuration, to conform to air transportability requirements. In addition, the weight of the compressor shall not exceed 20,000 pounds per tandem axle with the fuel tank three-quarters full in a reduced configuration. Achieving a reduced configuration shall be limited to the removal or relocation of mechanically attached (non-welded) components and shall not affect the transportability of the item, including the ability to negotiate, without interference, a 15-foot ramp at an angle of 17° between two horizontal surfaces. Removal and reinstallation or reinstallation time of all components required to achieve the reduced configuration shall not exceed four manhours. When delivered to the Government, the compressor shall not be in the reduced configuration.

3.17 Lubrication. Means for lubricating shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall be so located as to make the lubricating points easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery.

3.18 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill

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scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based or polyurethane enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.18.1 Air Force compressor. When specified for Air Force use (see 6.2), compressors shall be cleaned, treated, and painted in accordance with Air Force Drawing Number 7545352.

3.19 Lifting and tiedown attachments. The compressor shall be provided with a lifting eye for handling by a crane or a hoist. The eye shall be located at the top of the machine at the normal center of balance. The lifting eye shall be capable of supporting 2-1/2 times the weight of the machine and shall be mounted in such a manner that the machine and its accessories retain alignment during lifting operations. The compressor shall be equipped with tiedown attachments which conform to type II, or type III of MIL-STD-209. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the compressor on the carrier when shipped.

3.20 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be treated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.21 Name plate. A name plate will be furnished by the contracting officer for each compressor. The contractor shall stamp all necessary data in the blank spaces provided for that purpose and securely affix it to each compressor in a conspicuous place with non-ferrous metal screws or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

3.22 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with the manufacturer's standard practice.

3.23 Workmanship.

3.23.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to

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the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the desing. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.23.2 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

3.23.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, self-locking nuts, and screws shall be tight.

3.23.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. Welders shall be qualified, for the specified type of welding being performed, in accordance with applicable welding codes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification

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shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.5).

4.2.1 First article inspection. The first article inspection shall be performed on one complete compressor when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4.1, 4.4.2, 4.4.4, and when applicable, 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. Each compressor furnished under a contract shall be examined in accordance with 4.3 and tested in accordance with 4.4.3.

4.3 Examination. Each compressor shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. For the receiver, the compressor manufacturer shall have responsibility for verifying that the receiver has been inspected in accordance with the ASME Code. Official ASME Code symbols will be accepted as evidence of compliance.

4.4 Tests. Prior to the tests specified herein which require the compressor(s) be operated, the compressor(s) shall be serviced with oils and greases recommended by the manufacturer for use in the ambient temperature at the test location. Battery(s) used for test purposes shall be supplied by the manufacturer in lieu of the battery(s) to be shipped with the compressor.

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4.4.1 Performance test. The compressor shall be tested to determine capacity and power consumption in accordance with the rules, procedures, and instrumentation specified in the ASME PTC-9. The duration of the test shall be sufficient to record all data required under Report Test of PTC-9, but in no case shall the test duration be less than one hour. Test results shall be corrected to the atmospheric conditions specified in 3.6. Prior to the actual performance test, preliminary tests shall be run, as required, to verify that the compressor and associated system are in suitable condition, that speed controls have been properly adjusted, and that gages and instruments are functioning properly. Following the successful completion of the performance test, the compressor shall be operated for a period of not less than 30 minutes in each of the four tilted positions specified in 3.6. Inability of the compressor to meet the performance requirements of this specification or to operate satisfactorily in any of the four tilted positions shall be cause for rejection.

4.4.2 Functional test. The compressor shall be operated continuously for a period of not less than eight hours during which time a log shall be maintained of the capacity, discharge pressure, operating speed, and temperature of compressed air at the receiver discharge outlet. For a one-hour period during the test, the air discharge valve shall be alternately opened and closed so as to operate the compressor through the complete loading cycle (from zero to full rated capacity to zero capacity) not less than once every five minutes. The compressor shall operate at zero capacity for not more than one minute between cycles. Cycling operation shall be observed to determine if, when the air discharge valve is closed, the engine speed reduces from full-rated-capacity speed to a speed within the idling range recommended by the engine manufacturer. Failure of the compressor to meet the performance requirements of 3.6, 3.6.1, and 3.7, or malfunction or failure of any parts, components, or controls shall be cause for rejection.

4.4.3 Operational test. Each compressor, excluding first article samples, shall be tested for a period of not less than 30 minutes to verify satisfactory operation of the compressor and all controls. During the test all necessary adjustments in controls shall be made as required to insure that performance requirements relating to compressor output and regulation and engine speed will be met.

4.4.4 Mobility test. The compressor shall be towed over graded gravel roads for a distance of not less than 25 miles at speeds up to 20 mph, and over paved roads for a distance of not less than 25 miles at speeds up to 50 mph. After successful completion of the road test, the compressor shall be towed on off-road cross-country terrain for 10 miles at average speeds of approximately 5 mph. Off-road terrain is defined as open fields, ungraded roadways, and generally uneven terrain normally encountered at construction sites where compressors are used. During the mobility test, the compressor shall be checked for suspension capability and the parking brake shall be operated to verify that the brake will hold the unit on a 15 percent grade.

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4.5 Packaging inspection. The preservation, packing, and marking of the compressors shall be inspected to the requirements of section 4 of MIL-C-3600. When specified (see 6.2), a preproduction pack inspection shall be performed in accordance with MIL-C-3600.

5. PACKAGING

5.1 Preservation packing, and marking. The compressors shall be preserved, packaged, packed, and marked in accordance with MIL-C-3600. The level of preservation and packaging and level of packing shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not necessary.)

6.1 Intended use. The air compressors are intended as portable sources of compressed air. Each compressor is a self-contained unit readily movable and suitable for field service at construction sites.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Size of unit required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.2 and 4.2.1).
e. When a wet charged or wet sealed maintenance free battery shall be furnished (see 3.8).
f. Type of air end, if different (see 3.11).
g. When a manual shutoff valve is required (see 3.13).
h. When the undercarriage shall be equipped with a rigid towing tongue with adjustable hitch (see 3.15.1).
i. Type of tire, if different (see 3.15.3).
j. When a parking brake is not required (see 3.15.4).
k. Color required (see 3.18).
l. If, for Air Force acquisitions, cleaning, treatment, and painting shall conform to MIL-C-83286 (see 3.18.1).
m. When fungus treatment is required (see 3.20).
n. When a preproduction pack inspection is required (see 4.5).
o. Level of preservation and packing required (see 5.1).

6.3 Definitive specification part number. The specification part number is a definitive part number which corresponds to the sizes of compressors covered by this specification and defines the requirements of the options presented under this specification. The specification number, and the size code number are combined to form the definitive specification part number.

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6.3.1 Cataloging data. For cataloging purposes, part numbers for the compressors are assigned as follows:

	M82086	XX
Specification part number		
Class and size code number (see 6.3.2)		

6.3.2 Class and size. The class and size of the compressor are identified by a two-digit number (see table II).

TABLE II. Code letter to class and size.

Size	Class A	Class B
125	01	06
250	02	07
365	03	08
600	04	09
750	05	10

6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.5 Standard lubricants. When requested by the manufacturer, the contracting officer should obtain through established channels, a list of standard military lubricants applicable to the equipment covered herein. This information is contained in the Department of Defense Section of the Federal Supply Catalog, 9100-IL.

6.6 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory (see 6.2) as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.7 Changes in classification. The classification of compressors in this document differs from the superseded specification in the following respects:

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MIL-C-82086B	MIL-C-82086C
Class A - Skid mounted.	Class A - Wheeled undercarriage with fixed tongue.
Class B - Wheel-mounted.	Class B - Wheeled undercarriage with automotive steering and hinged towbar.

6.8 Subject term (key word) listing.

Rotary compressor
Undercarriage

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - YD
Air Force - 99

User activity:
Navy - MC

Preparing activity:
Navy - YD

Project 4310-0173

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-82086C	MILITARY SPECIFICATION - COMPRESSOR, AIR, DIESEL-ENGINE DRIVEN, PORTABLE, WHEEL-MOUNTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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